SOFTWARE TESTING PRACTICAL REPORT

TUTORIAL – 15
Automated Login Testing

Selenium WebDriver • Python • Authentication Validation • Assertions • Exception Handling

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Practical Title	Tutorial 15 — Automated Login Testing
Application Under Test	SauceDemo
Programming Language	Python
IDE / Automation Tool	PyCharm / Selenium WebDriver
Browser	Google Chrome
Test File	tutorial15.py

Practical aim

To create automated login functionality, authentication validation, assertions, and exception-handling scenarios using Selenium WebDriver with Python.

Prepared from the supplied Tutorial 15 lab manual, Python script, and screenshots. Screenshot evidence is described accurately; steps not visibly demonstrated are not marked as independently verified.

1. Aim and Objective

The aim is to test SauceDemo authentication using Selenium WebDriver. The script checks valid login, invalid password, invalid username, empty credentials, authentication assertions, error-message validation, exception handling, and logout.

2. Problem Statement

Create a Python Selenium script that automatically executes the five login-related scenarios specified in the lab manual and validates expected behavior through explicit waits and assertions.

3. Test Environment

Parameter	Details
Project	Automated Login Testing
Target URL	https://www.saucedemo.com/
Application module	Authentication / Login
SDLC model	Agile (as listed in the lab manual)
Language / IDE	Python 3.x / PyCharm
Automation library	Selenium WebDriver
Browser / driver	Google Chrome / Selenium-managed Chrome WebDriver
Wait strategy	WebDriverWait with a 10-second timeout
Valid test data	standard_user / secret_sauce
Negative test data	wrong_password; wrong_user; blank credentials

4. Test Scenarios and Expected Results

Test ID	Scenario	Expected Result
TC-01	Valid login	Inventory / Products page opens and authentication assertions pass.
TC-02	Invalid password	Login is rejected and an appropriate error message appears.
TC-03	Invalid username	Login is rejected and an appropriate error message appears.
TC-04	Empty credentials	Validation message indicates that the username is required.
TC-05	Logout	User returns to the login page.

5. Web Element Identification

Element	Locator Type	Locator Value	Purpose
Username field	ID	user-name	Enter username
Password field	ID	password	Enter password
Login button	ID	login-button	Submit credentials
Error message	CSS Selector	h3[data-test='error']	Validate login error
Menu button	ID	react-burger-menu-btn	Open navigation menu
Logout link	ID	logout_sidebar_link	Log out
Products heading	Class Name	title	Verify successful login

6. Test Procedure / Automation Flow

1. Initialize Chrome WebDriver and maximize the browser.

2. Open SauceDemo and wait until the username, password, and login controls are ready.

3. For TC-01, enter valid credentials, submit the form, and verify the inventory URL and Products heading.

4. For TC-02, use a valid username with an invalid password and check the error message.

5. For TC-03, use an invalid username with a valid password and check the error message.

6. For TC-04, submit the form with both fields blank and validate the username-required message.

7. For TC-05, log in, open the menu, select Logout, and verify the login button returns.

8. Close each browser instance in a finally block so it is cleaned up even if an error occurs.

7. Test Case Details

Field	Details
Test Case ID	TC-15-01 (valid login)
Requirement ID	REQ-LOGIN-01
Module / feature	Authentication / Login
Test objective	Verify successful authentication with valid credentials
Test type / technique	System testing / Black-box testing
Priority / severity	High / Critical (as listed in the lab manual)
Precondition	SauceDemo is reachable and Chrome WebDriver is configured
Test data	standard_user / secret_sauce
Expected result	Products / inventory page appears
Execution status	See evidence and execution notes in this report

8. Assertion and Exception Handling

Assertions in the supplied script: the URL should contain “inventory”; the Products heading should be visible (per lab-manual reference implementation); invalid credentials should produce the expected error text; empty credentials should display “Username is required”; logout should return to the login screen.

Exception	Possible Cause	Handling in Script
NoSuchElementException	A locator cannot find an element	Caught in the valid-login test
TimeoutException	An element or URL condition is not met within the wait period	Caught in relevant test cases
ElementNotInteractableException	An element cannot be interacted with	Caught in the valid-login test
StaleElementReferenceException	The DOM changes after locating an element	Caught in the valid-login test
AssertionError	Actual result does not match the expected result	Caught to print a FAIL message

9. Source Code Summary

The submitted code defines start_browser() to initialize Chrome and login(driver, username, password) to enter credentials. Five separate test functions cover valid login, invalid password, invalid username, empty credentials, and logout. Each function manages its own browser and uses explicit waits to reduce timing-related failures.

Execution-time note: the supplied script calls time.sleep(60) in each test’s finally block. Since five tests run sequentially, this adds at least five minutes of fixed waiting. For routine test runs, remove the fixed sleeps or replace them with condition-based waits; keep screenshots or manual inspection as a separate optional step.

10. Representative Source Code

The excerpt below summarizes the shared setup and login helper from the supplied script. The complete script provided by the student contains all five test functions.

```
URL = "https://www.saucedemo.com/"
VALID_USERNAME = "standard_user"
VALID_PASSWORD = "secret_sauce"

def start_browser():
driver = webdriver.Chrome()
driver.maximize_window()
return driver

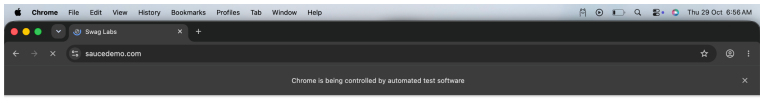
def login(driver, username, password):
driver.get(URL)
wait = WebDriverWait(driver, 10)
username_box = wait.until(
EC.visibility_of_element_located((By.ID, "user-name"))
)
password_box = wait.until(
EC.visibility_of_element_located((By.ID, "password"))
)
login_button = wait.until(
EC.element_to_be_clickable((By.ID, "login-button"))
)
username_box.clear()
username_box.send_keys(username)
password_box.clear()
password_box.send_keys(password)
login_button.click()

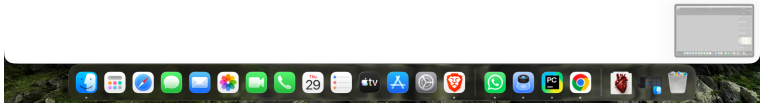
# Each test uses try/except/finally:
# TC-01 valid login; TC-02 invalid password;
# TC-03 invalid username; TC-04 empty credentials;
# TC-05 logout. The browser is closed with driver.quit().
```

11. Screenshot Evidence

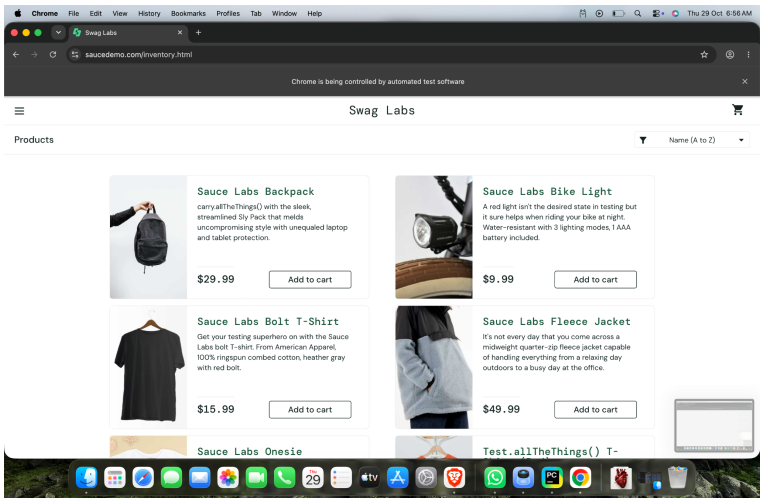
The following screenshots were supplied. They show the SauceDemo loading state, the Products inventory page, a Google Password Manager warning layered over the inventory page, and console output. They do not visibly show all four negative-login outcomes or the logout result.

Evidence 1 — SauceDemo browser loading state. This image alone does not confirm a completed login.

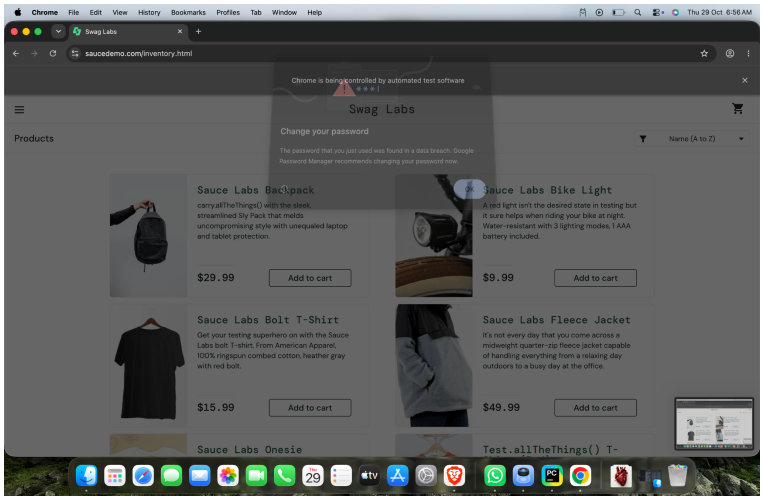




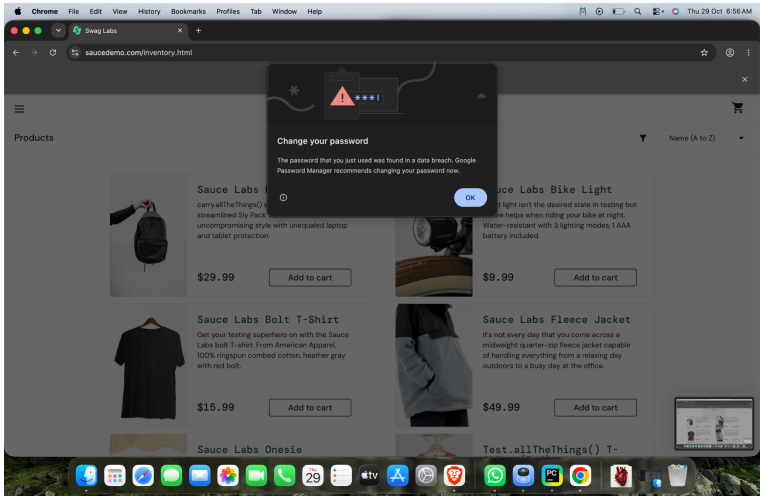
Evidence 2 — SauceDemo Products inventory page, indicating that the inventory page was reached.



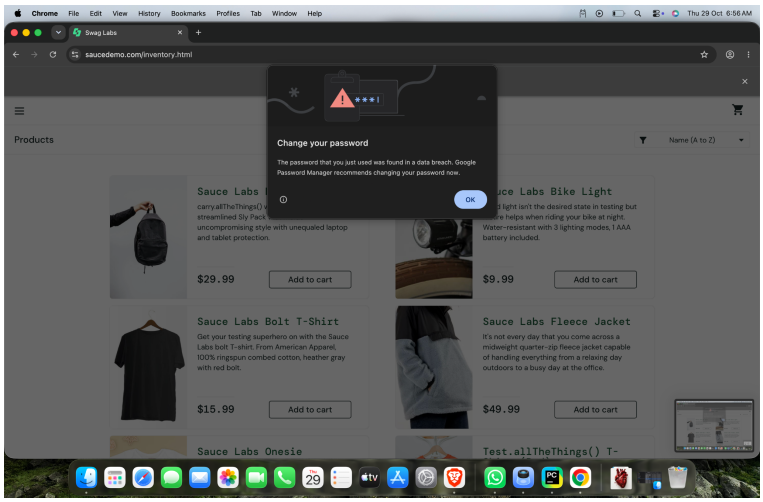
Evidence 3 — Inventory page dimmed by a browser password warning.



Evidence 4 — Google Password Manager warning stating the recently used password was found in a data breach.



Evidence 5 — Repeated password warning screenshot; this is a browser-level prompt, not a SauceDemo authentication error.



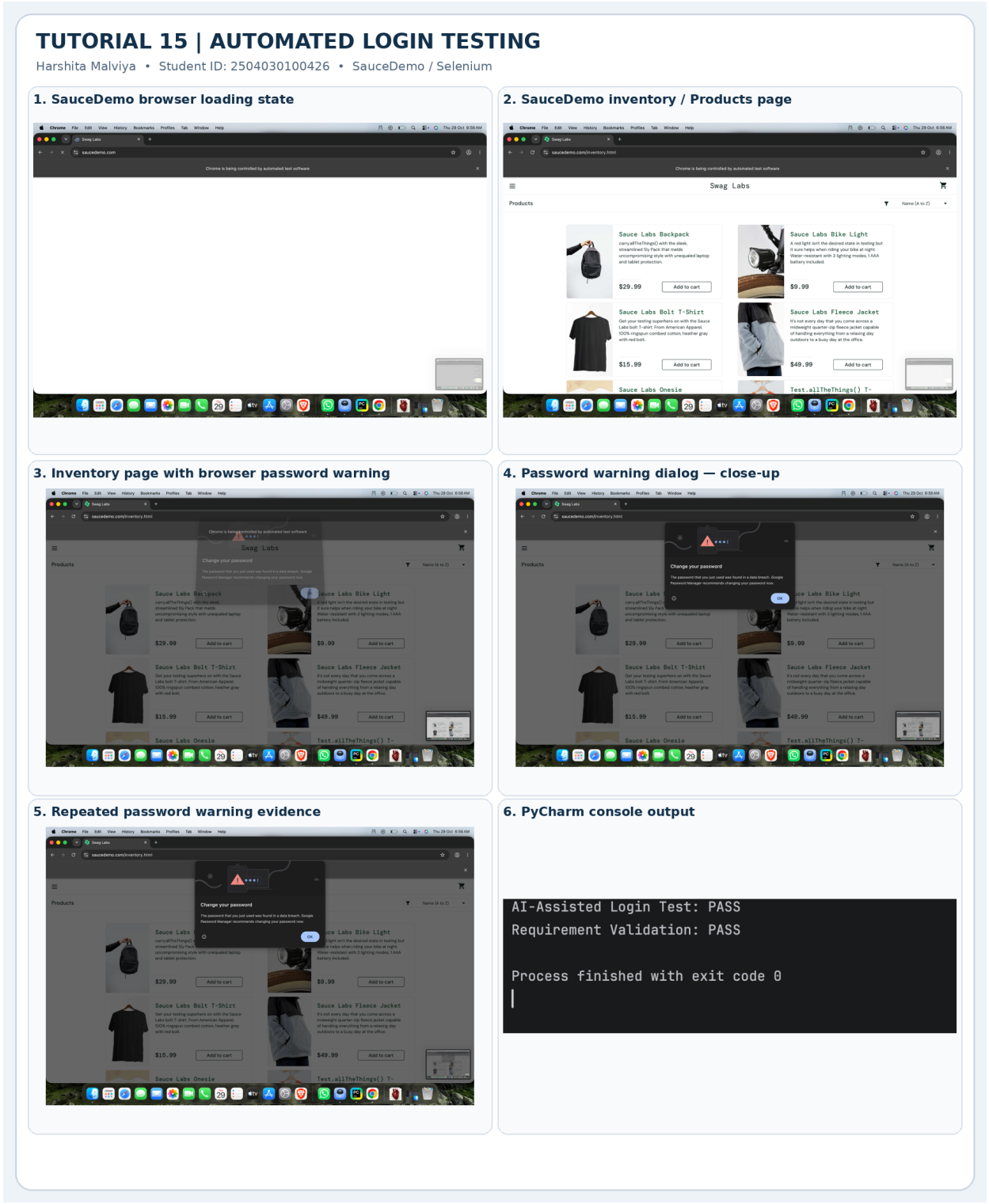
Evidence 6 — Console output shows “AI-Assisted Login Test: PASS”, “Requirement Validation: PASS”, and exit code 0. This is related execution evidence, not output for all five test functions shown in the submitted script.

```
AI-Assisted Login Test: PASS
Requirement Validation: PASS

Process finished with exit code 0
```

12. Screenshot Collage

The collage groups the six screenshots supplied for Tutorial 15 into one evidence sheet. The browser password-warning images are identified as browser-level prompts rather than application error messages.



13. Execution Summary and Evidence Status

Item	What is supported by supplied evidence
Browser / application	SauceDemo domain and inventory page are visible in browser screenshots.
Successful login	Inventory page is visible, which is consistent with successful login.
Invalid password	The exact SauceDemo invalid-password error is not visible in supplied screenshots.
Invalid username	The exact SauceDemo invalid-username error is not visible in supplied screenshots.
Empty credentials	The username-required validation message is not visible in supplied screenshots.
Logout	The post-logout login screen is not visible in supplied screenshots.
Console	A related console screenshot shows PASS for an AI-assisted login test and requirement validation, plus exit code 0; it does not list TC-01 through TC-05.

14. Observations

1. The inventory screenshot shows the SauceDemo Products page.
2. The password-warning dialog comes from Google Password Manager and should not be confused with a login error generated by SauceDemo.
3. The console screenshot shows a successful process exit for a related test run, but it does not prove every test case in the submitted script passed.
4. The report therefore distinguishes visible evidence from expected behavior described in the code and lab manual.

15. Result and Conclusion

The submitted Tutorial 15 script is structured to test valid authentication, invalid credentials, empty-field validation, assertions, exception handling, and logout on SauceDemo using Selenium WebDriver. The supplied screenshots confirm that the inventory page was reached and show related console PASS messages. To document a complete run, capture the console output listing TC-01 through TC-05 and screenshots of the three negative-login messages and the post-logout login page.

16. Questionnaire

Q1. What is authentication testing?
Authentication testing checks whether valid users can access an application and invalid users are rejected.

Q2. What is Selenium WebDriver?
It is an API used to automate and control web browsers.

Q3. What is an assertion?
An assertion compares an expected result with the actual result to determine whether a test condition passes.

Q4. What is exception handling?
It is a programming technique used to handle runtime errors without abruptly stopping the program.

Q5. Which exception occurs when an element cannot be found?
NoSuchElementException.

17. Recommended Final Evidence Checklist

■ PyCharm source code ■ Login page ■ Valid login / Products page ■ Invalid password message ■ Invalid username message ■ Empty-credentials message ■ Logout result ■ Console output listing each test-case status